

# PCB Lab

## 1. Initial Parameters

$$N_1 = 4, N_2 = 1, N_3 = 2$$

$$A_0 = 5 + \frac{N_1}{2} = 7 \text{ dB}$$

$$f_l = 10(5 + \frac{N_2}{2}) = 55 \text{ Hz}$$

$$f_h = 1000(10 + N_3) = 12000 \text{ Hz}$$

$$\text{frequency cutoff} = 7 \text{ dB} - 3 \text{ dB} = \mathbf{4 \text{ dB}}$$

## 2. Calculations

$$V_b = \frac{5}{R_1} \div \left( \frac{1}{R_1} + \frac{1}{R_2} \right) = 1.67 \text{ V}$$

$$V_{be} = 0.7 \text{ V} = V_b - V_e \rightarrow V_e = 0.97 \text{ V}$$

$$I_e = \frac{V_e}{R_e} = 4.85 \text{ mA}$$

$$I_c = \alpha I_e \approx I_e$$

$$g_m = \frac{I_c}{V_T} = 0.194 \frac{\text{A}}{\text{V}}$$

$$r_e = \frac{I_e}{V_T} = 5.155 \Omega$$

$$\Delta V_i = i_e(r_e + R_e)$$

$$\Delta V_o = -i_c(R_c)$$

$$A_{dB} = 7 \text{ dB} = 20 \log(A_o) \rightarrow A_o = -2.2387 \frac{\text{V}}{\text{V}}$$

$$\frac{\Delta V_o}{\Delta V_i} = -\frac{\alpha R_c}{r_e + R_e} = A_o \rightarrow R_c = 459.28 \Omega$$

$$C_b = \frac{1}{2\pi f_l(R_1 \parallel R_2)} = 4.34 \mu\text{F}$$

$$C_o = \frac{1}{2\pi f_h(R_c)} = 28.88 \text{ nF}$$

### 3. Screenshots

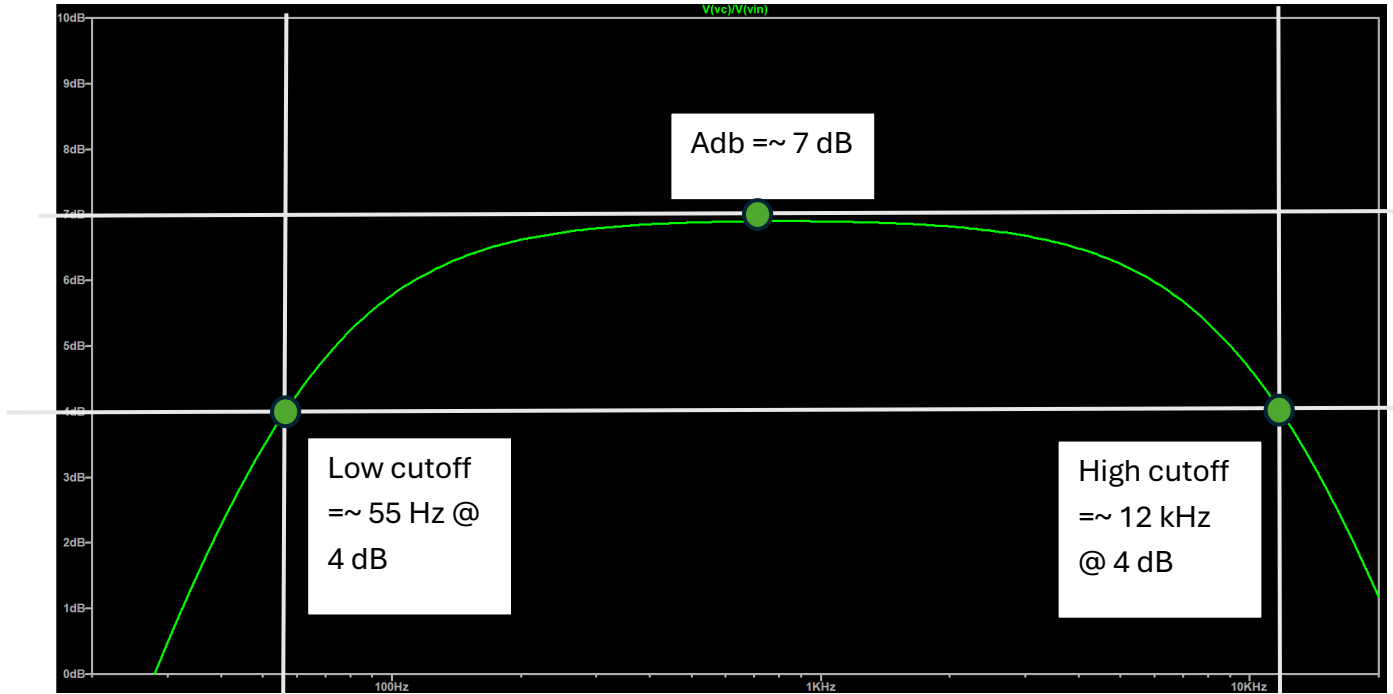


Figure 1: Decade plot showing gain, marking low and high frequency cutoffs and max gain.

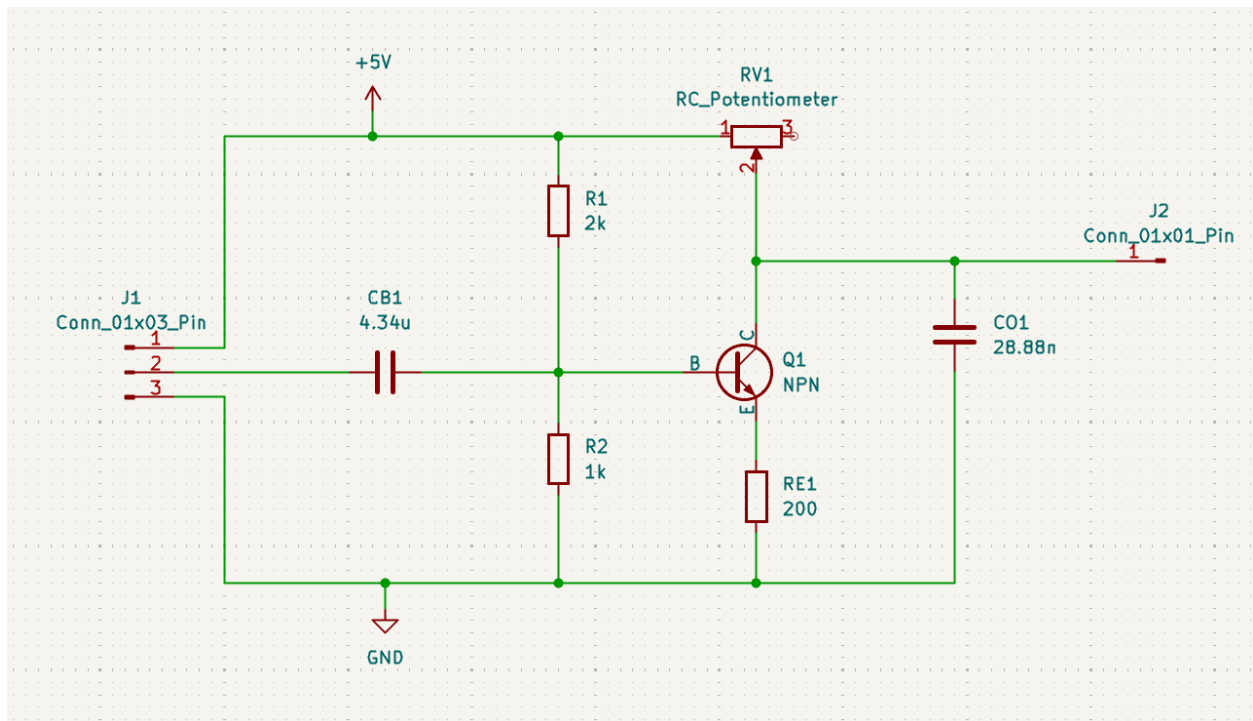


Figure 2: Schematic using calculated resistance and capacitance values for initial parameters.

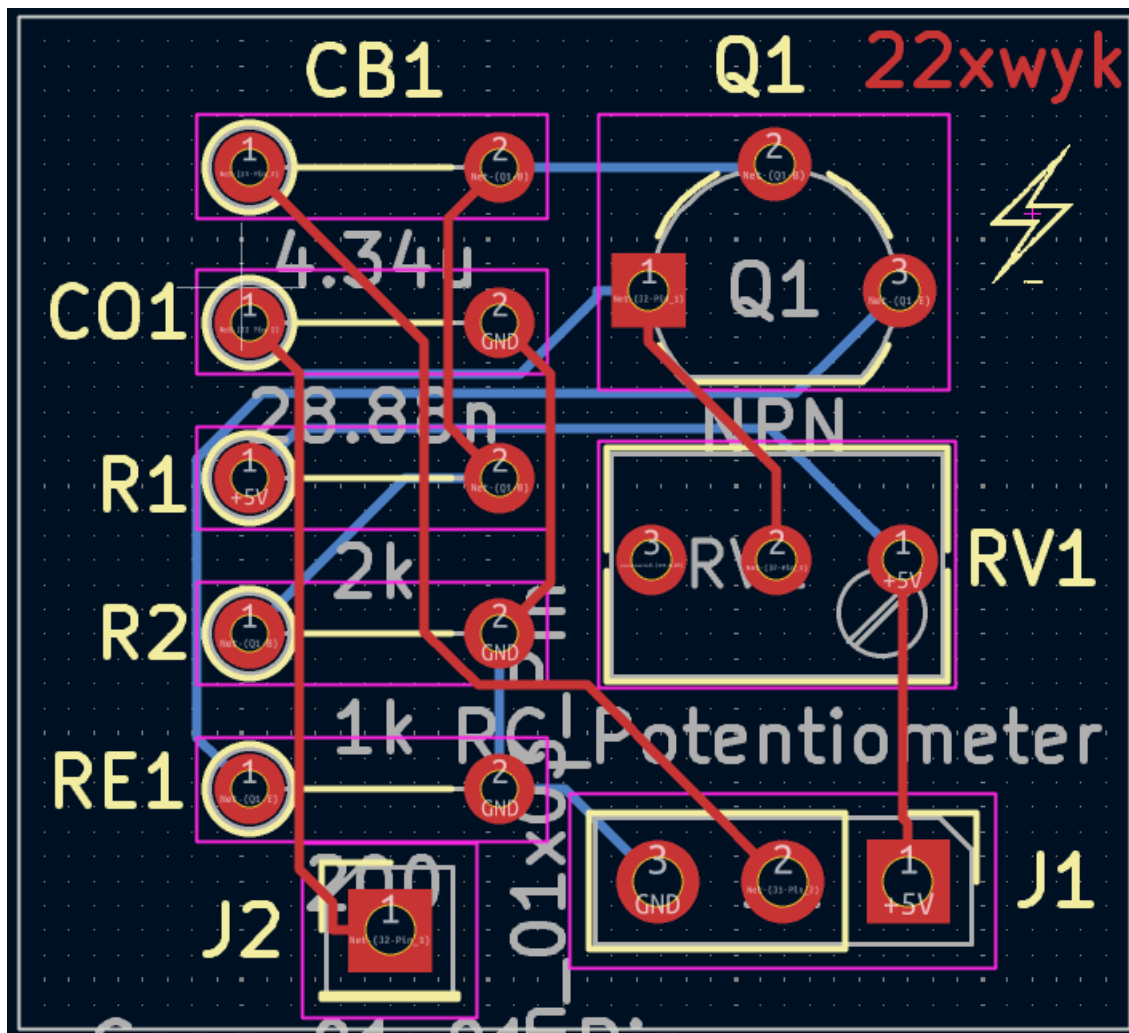


Figure 3: Screenshot of PCB layout, showing components and routing.